

calB
| L.H.R.W.
M.A.C.L.S.
63/4

ABERDARE TECHNICAL COLLEGE
P. O. BOX 1815 - 10101 KARATINA TEL: 0726507444
E-MAIL: INSTITUTE.ABERDARE@GMAIL.COM

APPLY BASIC ELECTRONICS

INSTRUCTIONS TO CANDIDATE

1. This paper consists of **TWO** sections: A and B.
2. Attempt **ALL** questions in section A and **ANY THREE (3)** questions in section B.
3. Marks for each question are indicated in the brackets.
4. Do not write on the question paper.

18/11

Features	Smartphones	Computers
a) Central Processing Unit - (CPU)	Uses one	uses one.
b) Memory RAM	Has RAM	Has RAM
c) Storage	Have internal storage SSD/ROM	Have internal storage SSD/HDDs
d) Operating System	Has sophisticated OS Android/iOS	Has sophisticated OS Windows, Linux
e) Input output interfaces	Touchscreen keyboard, sensors and has speakers.	Has Keyboard and speakers.

Explain and table!

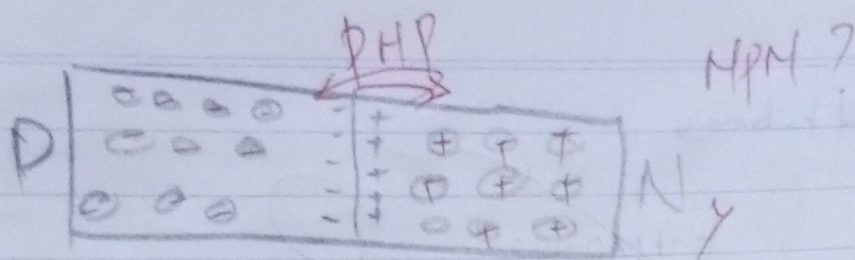
b) Power supply ~~regulator~~ and voltage regulation. → The circuit must have a stable and efficient power supply.

Signal integrity and interconnections. → Proper layout of traces and connectors is crucial to minimize noise and signal loss.

Component selection and thermal management. Components must be selected that meet performance and ensure they manage heat dissipation in motherboards.

C.) Firmware and Software Vulnerabilities

- Production system disruption. (2)
- Intellectual Property Theft. explain.
- Supply Chain attacks.



2. Semiconductors memory are electronic or silicon based and stores data using electrical charges while optical memory uses laser light on reflective surfaces. (2)

3. Two batteries = 1.5V

$$R = 0.5\Omega = 0.5\Omega \div 2 = 0.25\Omega$$

$$\text{Bulb } 6\Omega + 6\Omega = 12\Omega$$

$$R_{\text{eq}} = R_1 + R_2 = 6 + 6 + 0.25 = 12.25$$

$$C = \frac{V}{R}$$

$$= \frac{1.5}{12.25}$$

$$= 0.122A$$

4. Signal diodes $\rightarrow$ They allow current flow in ~~one~~ ^{specific} direction only. (3)
Capacitors $\rightarrow$ Stores electrical energy (1)

5. When temperature increases more charge carriers are generated increasing conductivity. (2)

6. 135_{10} to binary.

$$1_{10} = 0001_2$$

$$= 100110101_2$$

$$3_{10} = 0011_2$$

$$5_{10} = 0101_2$$

~~126~~ 126_{10} to hexadecimal

number result remainder

126

7

14

$$= 7E_{16}$$

7

0

7

1001001_2 to decimal

$$(1 \times 2^6) + (0 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$1 + 0 + 0 + 8 + 0 + 0 + 1 = 10$$

$$= 73_{10}$$

1434 octal to binary

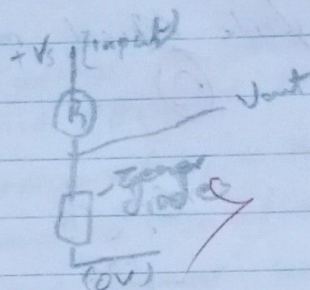
1 = 001_2

4 = 100_2

3 = 011_2

4 = 100_2

$E = 1100011100_2$



7.

8. Junction diodes are used for converting AC to DC and also prevent the flowback of charges.

9. The bulb kept on lighting even after closing the switch. meaning rc effect

10. Valence band are the highest occupied energy band while conduction band is the lowest occupied energy band.

11 Types of ROM

~~PROM~~ PROM (Programmable ROM)

EPROM (Erasable Programmable ROM)

EEPROM (Electrically Erasable Programmable ROM)

12 $C_T = C_1 + C_2$

$10\mu F + 0.30\mu F + 2.5\mu F = 12.8\mu F$

$12.8\mu F = 12.8 \times 10^{-6} F / 0.000025 F$

Charge Q

$Q = CV$

$Q = 0.0000128$

$0.0000128 \times 24 = 0.0003072 C$

13 They are less affected by noise and distortion.

They are easy for integration and more reliable.

4. As the forward bias voltage increases depletion layer narrows allowing large current to flow easily beyond point A.

Section B.

15) PN junction diode: This is junction diode fabricated semiconductor device

formed by joining P and N type semiconductor materials

→ Made of formation → One side is doped with acceptor impurities while other side doped with donor impurities.

→ Acts as a rectifier converting AC to DC.

→ Application: Used in power supply units converting AC to DC like in chargers.

2. Light emitting diode (LED)

This is special PN junction that emits light visible when forward biased.

Made of formation → formed when forward biased, electrons & holes recombine at the junction releasing energy in form of photons (light).

Converts electrical energy directly to light energy.

Used in display screens.

b. Access control and Authentication → This is by use of strong user authentication and role based access control.

Data encryption and backup strategies → Always ensure to encrypt data whether stored or in transit.

Use firewalls and intrusion detection systems and Antivirus software.

This is to monitor and block any suspicious traffic.

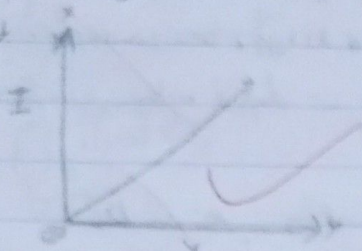
c. Simplicity and Reliability of hardware → electronic components can reliably represent and switch between the (off or on (0 and 1)) quickly

Ease of logical operations All the computer operations can be performed using simple boolean logic gates.

Ohm's Law states that Current flowing through a conductor is directly proportional to the potential difference across its ends.

Voltage method: Keep resistance constant and adjust the voltage using power supply and record corresponding current readings.

Resistance method: Keep Voltage constant and adjust the resistor to change resistance and record current.



35₁₀ - 17₁₀ to decimal notation

$$35_{10} = 00100011_2$$

$$17_{10} = 00010001_2$$

To calculate 35 - 17, invert 17 and add 1 to 11101111

$$\begin{array}{r} 00100011 \\ + 11101111 \\ \hline 0000010_2 \end{array} = 18_{10} = 00010010_2$$

$$17_{10} = 2^4$$

$$25_6 = 11100_2$$

$$\text{invert } 17 = 01110$$

$$11100 = 01011_2$$

$$1$$

$$17_{10} = 10001_2$$

$$+ 01110$$

$$= 11_{10}$$